

Silverio Martínez Andrés

Álgebra Lineal

Grupo 17

Semestre 2021-2

Tarea 13

Para el espacio vectorial complejo $\mathbb{C} = \{z \mid z = a + bi, i^2 = -1; a, b \in \mathbb{R}\}$ determinar si la función $F: \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$, definida por

$$F(z_1, z_2) = z_1 \overline{(z_2 + i)} \quad \forall z_1, z_2 \in \mathbb{C}$$

Es un producto interno. En caso afirmativo, calcular la norma del vector $z = 2 - 3i$; en caso contrario, indique los axiomas de la definición que no se cumplen.

Solución

1- Simetría-

$$(\vec{u} | \vec{v}) = \overline{(\vec{v} | \vec{u})}$$

$$(z_1, z_2) | (z_2, z_1) = \overline{(z_2, z_1) | (z_1, z_2)}$$

$$z_1 \overline{(z_2 + i)} = \overline{z_2 \overline{(z_1 + i)}}$$

$$z_1 (z_2 - i) = \overline{z_2 (z_1 - i)}$$

$$z_1 z_2 - z_1 i = \overline{z_2 z_1 - z_2 i}$$

$$z_1 z_2 - z_1 i \neq z_1 z_2 - z_2 i$$

$$(a+bi)(c+di) - (a+bi)i = (c+di)(a+bi) - (c+di)i$$
$$ac+adi+bc i+db i^2 - ai - bi^2 = ca+cb i+ai+db i^2 - ci - di^2$$
$$ac+adi+bc i-bd-ai+b = ac+bc i+ad-bd-ci+d$$

$\therefore$ No comple //

2.- Distributividad

$$z_1 = a + bi$$

$$z_2 = c + di$$

$$z_3 = e + fi$$

$$(\bar{0} | \bar{v} + \bar{w}) = (\bar{0} | \bar{v}) + (\bar{0} | \bar{w})$$

$$(z_1 | z_2 + z_3) = (z_1 | z_2) + (z_1 | z_3)$$

$$z_1 | (\overline{z_2 + z_3 + i}) = (\overline{z_1 | z_2 + i}) + (\overline{z_1 | z_3 + i})$$

$$z_1 | (\overline{z_2 + z_3 - i}) = (\overline{z_1 | z_2 - i}) + (\overline{z_1 | z_3 - i})$$

$$z_1 z_2 + z_1 z_3 - z_1 i \neq z_1 z_2 - z_1 i + z_1 z_3 - z_1 i$$

$z_1 z_3 \neq z_1 i$ $\therefore$ No comple //

3.- Homogeneidad

$$(\alpha \bar{0} | \bar{v}) = \alpha (\bar{0} | \bar{v})$$

$$(\alpha z_1 | z_2) = \alpha (z_1 | z_2)$$

$$\alpha z_1 | (\overline{z_2 + i}) = \alpha (\overline{z_1 | z_2 + i})$$

$$\alpha z_1 | (\overline{z_2 - i}) = \alpha (\overline{z_1 | z_2 - i})$$

$$\alpha z_1 z_2 - \alpha z_1 i = \alpha (z_1 z_2 - z_1 i)$$

$$\alpha z_1 z_2 - \alpha z_1 i = \alpha z_1 z_2 - \alpha z_1 i$$

$$z_1 = a + bi$$

$$z_2 = c + di$$

$$\alpha (a + bi)(c + di) - \alpha i = \alpha (a + bi)(c + di) - \alpha (a + bi)i$$

$$\alpha (ac + adi + bci + bdi^2) - \alpha (ai + bi^2) = \alpha (ac + adi + bci + bdi^2) - \alpha (ai + bi^2)$$

$$\alpha ac + \alpha adi + \alpha bci - \alpha bd - \alpha ai + \alpha b = \alpha ac + \alpha adi + \alpha bci - \alpha bd - \alpha ai + \alpha b$$

Si comple //

4. Positividad

$$(\bar{u} | \bar{v}) > 0 \quad \bar{u} \neq 0$$

$$(z_1 | z_1) > 0$$

$$z_1 (\overline{z_1 + i}) > 0$$

$$z_1 = a + bi$$

$$z_1 (z_1 - i) > 0$$

$$z_1^2 - z_1 i > 0$$

$$(a + bi)^2 - (a + bi)i > 0$$

$$a^2 + 2abi + b^2 i^2 - ai - bi^2 > 0$$

$$a^2 + 2abi - ai - b^2 + b > 0$$

$$a = -bi$$

$$(-bi)^2 + 2(-bi)(bi) - (-bi)i - b^2 + b > 0$$

$$(b^2 i^2) + 2(-b^2 i^2) - (-bi^2) - b^2 + b > 0$$

$$-b^2 + 2(b^2) - (b) - b^2 + b > 0$$

$$-2b^2 + 2b^2 - b + b > 0$$

$$0 = 0$$

o.o X No cumple

$\therefore F: \mathbb{C} \times \mathbb{C} \rightarrow \mathbb{C}$ No es un producto interno, ya que no se cumplen 3 axiomas (Simetría, distributividad y positividad)